MILITARY SPECIFICATION

RING, SHEAR SEAL
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for shear seal rings used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

11829083	Ring, Shear Seal

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on the date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B-29	Pig Lead

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Construction. The shear seal ring shall comply with all requirements specified on drawing (dwg.) 11829083, with all requirements specified herein and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

3.2 Materials. The contractor shall provide objective evidence (see 6.4) that materials comply with applicable dwgs. and specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

3.3 First article inspection. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.4 Workmanship. The shear seal ring shall be free of distortions, deep gouges, nicks or dents, or any other obvious physical defects. Parts shall be clean and free of dirt, grease or any other foreign matter. Cleaning methods shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use is own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

1. First Article Inspection (see 4.3)
2. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.3), the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Ring Shear Seal	11829083	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg. and to the tests and inspections specified in the first article inspection table herein. All parts shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract and all materials shall be obtained from the same source of supply as will be used in regular production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

FIRST ARTICLE INSPECTION

MIL-R-63166(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.1	Ring, Shear Seal				11829083
					NEXT HIGHER ASSEMBLY: M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR /100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classification of Defects and Tests	6		3.1, 3.2 3.4	4.4.2.1

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots of shear seal rings shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examinations. See MIL-A-48078 and the Quality Conformance Inspection Tables herein.

(a) Major defects: Unless otherwise specified, inspection for Major defects shall be sampled in accordance with Table I.

(b) Minor defects. Unless otherwise specified, inspection for Minor defects shall be sampled in accordance with Table II.

Table I. Sampling plan for major defects.

LQ = .05 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table II. Sampling plan for minor defects.

LQ = .10 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

MIL-R-63166(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Ring, Shear Seal				11829083
					NEXT HIGHER ASSEMBLY
					M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Outside diameter		Table I	3.1	Gage
102.	Large inside diameter		Table I	3.1	Gage
103.	Location of large inside diameter		Table I	3.1	Gage
104.	Length of large inside diameter		Table I	3.1	Gage
105.	Small inside diameter		Table I	3.1	Gage
106.	Location small inside diameter		Table I	3.1	Gage
107.	Overall length		Table I	3.1	Gage
108.	Angle of inside bevel		Table I	3.1	Gage
109.	Length of outside diameter		Table I	3.1	Gage
110.	Part identification (dots)		Table I	3.1	Visual
111.	Material per ASTM B29		-	3.2	Visual (see note below)
Minor:					
201.	Poor workmanship		Table II	3.4	Visual

NOTES: A certificate of analysis will be required verifying the chemical composition of a representative sample of each lot of lead using the wet chemical method per ASTM B29.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Packaging				See 5.
					NEXT HIGHER ASSEMBLY: N.A.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging and packing not in accordance with applicable requirements		-	5.0	Visual
102.	Objective evidence (see 3.2) not available for packaging and packing material		-	5.1.1	Visual
103.	Parts not clean and free of dirt and other foreign material prior to packaging		100%	5.1.1	Visual
104.	Parts not individually packaged		100%	5.1.1	Visual
105.	Stiffening and cushioning of packages inadequate to provide firm package (when required)		100%	5.1.1	Visual/Manual
106.	Individual packages not properly heat sealed		100%	5.1.1	Visual
107.	Intermediate pack improper or not properly sealed		100%	5.1.2	Visual/Manual
108.	Exterior pack improper or not properly sealed		100%	5.1.3	Visual
109.	Marking of individual packages, intermediate packs or exterior packs missing, improper or illegible		100%	5.1.4	Visual
110.	Evidence of poor workmanship		100%	-	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see section 6 of MIL-A-48078 and 6.3 herein).

4.5 Test methods. Not applicable.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract or order, packaging and packing shall be in accordance with MIL-STD-1169 subject to the following details and conditions.

5.1.1 Packaging. Items shall be single item packaged in fiberboard containers, sealed in waterproof barrier bags and heat sealed in accordance with MIL-STD-1169, Level A. Barrier bags shall be fabricated from MIL-B-121 (Type II, Grade A, Class 1) material or an approved equivalent. Fillers, positioners, stiffeners and cushioning adequate to prevent physical damage to the item and to finished surfaces during transit, storage and handling shall be provided as required.

5.1.2 Intermediate packing. Single item packages shall be packed in intermediate packages in accordance with MIL-STD-1169, Level B.

5.1.3 Packing. Intermediate packages shall be packed in accordance with MIL-STD-1169, Level B.

5.1.4 Marking. Unit packages, intermediate packs and exterior containers shall be marked with the following:

a. Item nomenclature
b. Part or dwg. no.
c. Used on M753 Projectile
d. Manufacturer
e. Quantity enclosed
f. Lot no.
g. Contract or purchase order no.
h. Date of acceptance

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. Shear seal rings covered by this specification are intended for use on Motor Body Bonding Assemblies of rocket motors for M753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article inspection, when required (see 3.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander U.S. Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Objective evidence. A statement (certification) supported by test data that each lot of material meets the requirements of the applicable dwgs. and specifications.

6.5 Supersession data. This specification includes the requirements of Purchase Description AR-PD-20 dated 9 April 1979 with Amendment 1 dated 6 March 1981.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

6.6 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:

Army-AR

Preparing activity

Army-AR

PROJECT NUMBER: 1110-A169

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.